

Itzhak Bars

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Overview

My current interests include mainly String Field Theory ([SFT](#)), and Two-Time Physics ([2T-Physics](#)), but these are related to several other topics in which I am currently active as outlined below.

Some of my [early contributions to string theory](#) include the non-trivial quantization of the two-dimensional string, and the first treatment of 1-branes interacting with 0-branes (interpreted as quarks at the ends of a string). This work established a firm and explicit relation between [string theory and large-N](#)

QCD in two dimensions. The more recent AdS-CFT correspondence includes an analogous endeavor that establishes a relation between field theory and string theory in one higher dimension. The AdS-CFT approach mainly involves the gauge sector, as opposed to the matter sector treated in my early work. How to make use of such ideas to make progress in real QCD in four dimensions is one of the challenges that I think about.

The action for the superparticle suggested in 1981 by Brink & Schwarz was discovered earlier in 1975 in the context of supersymmetric quarks at the ends of a string. The covariant quantization of the superparticle and superstring continue to be a challenge today, and is part of my research on the way to the supersymmetric generalization of string field theory.

In the conformal field theory era of string theory I emphasized the importance of strings moving in backgrounds with curved space-time, since string theory should play its main role during the (surely curved) early universe. With this in mind, I introduced some of the first exactly solvable string models in curved spacetime, including the $SL(2,R)/R$ model which was later understood by Witten to be a model for strings on black holes. I continue to research this general topic in the context of string field theory and cosmology.

My interest in higher dimensions began by providing the first evidence that 11 dimensions is critical for the quantum consistency of the supermembrane. After the strings-95 conference that marked the second superstring revolution, I emphasized in my 1995 talks that structures based in (10,2) dimensions were evident in the extended algebra of M-theory. Soon afterwards, papers suggesting more timelike dimensions began to appear. This led to my proposal of an algebraic S-theory in 1996, which helped develop some 2T notions in collaboration with Costas Kounnas. Finally that approach developed into a rather basic and fundamental dynamical form based on $Sp(2,R)$ gauge symmetry in a 1998 paper with Oleg Andreev and Cemsinan Deliduman, and then transformed into a general 2T-physics theory in its own right. Later I found connections to some work by Dirac in 1926 and Salam in the 1970's in connection with conformal symmetry $SO(d,2)$ (one extra space and one extra time compared to Lorentz symmetry $SO(d-1,1)$), which can now be understood as some of the consequences of 2T-physics. The sort of symplectic gauge

symmetry of 2T-physics may well explain the duality and holographic properties of the mysterious M-theory.

My work on 2T field theory based on the Moyal product is what led me to discover the Moyal star formulation of string field theory (MSFT). This was motivated by the observations that, on the one hand, a very similar formalism of non-commutative field theory and Chern-Simons type actions emerged naturally in both cases, and on the other hand, bosonized ghosts in string field theory play a role similar to a second timelike coordinate. Through MSFT I set

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String Field Theory (SFT)

Since the early stages of string theory in the 1970s, string field theory (SFT) was recognized to be a non-perturbative approach to string theory. Among all the formulations of string theory since its inception, string field theory stands out as the most complete scheme as a non-perturbative formulation that seems, in principle, to be better positioned to answer the central physics questions. One of the main goals of SFT (and indeed of all the efforts in string theory) is understanding the vacuum state and how we ended up in four dimensions. This includes elucidating the physics of the very early universe and how this determined the gauge symmetries (forces) and the families of quarks and leptons (matter) that we observe today.

Many attempts have been made over the past 30 years to formulate and develop computational tools in SFT. While all of these approaches were correct, the proposed formalisms were too cumbersome to extract easily non-perturbative or even perturbative information about string theory. In 2001 I introduced the Moyal Star Formulation of String Field Theory (MSFT) as a computational framework for Witten's cubic open string field theory. The novelty was that string interaction was re-formulated in terms of the simple Moyal product indicative of an induced quantum mechanics which is produced by string joining. The advantage of MSFT is that the resulting non-commutative field theory is much simpler for practical computations because the Moyal star product replaces conformal field theory or the oscillator approach in all SFT computations. This leads to new non-perturbative computational techniques

directly in MSFT without recourse to cumbersome maps to conformal field theory which is anchored essentially in perturbative string theory.

MSFT is now the simplest description of string interactions in the context of a complete and nonperturbative formulation of open string theory. Having demonstrated that D-branes, as well as closed strings, emerge non-perturbatively in open string field theory, it is quite possible that a supersymmetric version of MSFT (still to be achieved) will amount to some version of the complete M-theory.

Much of the new computational technology was developed with my collaborators Yutaka Matsuo and Isao Kishimoto, and more recently with Inyong Park. We have applied MSFT to both perturbative string physics (new results for off-shell string amplitudes) and non-perturbative string physics (analysis of the true vacuum of string theory, D-branes). In these areas MSFT has yielded new results that were not obtained before, while at the same time we have verified that MSFT is in detailed agreement with other approaches to string theory (conformal field theory, oscillator formalism etc.).

Finding the true vacuum is one of the most important challenges in string theory. We introduced an [analytic approach](#), and applied it to the solution of the classical string field equations, including interactions. It was possible to obtain all exact solutions, including the vacuum solution, when an anomalous term in the energy of the midpoint of the string is neglected. The anomalous midpoint energy was then treated as a perturbation, and the first two terms were computed. This ongoing program is expected to yield analytic insight into the true vacuum of string theory.

Ongoing projects include generalizing the MSFT approach to strings in curved backgrounds (I have in mind applications to early cosmology) and supersymmetrization in the covariant Green-Schwarz or Berkovits formalism (which will amount to a full definition of a version of M-theory). I have already achieved the first important step of formulating the generalization of the Moyal star product for strings in these conditions, and will next construct the BRST operator, which will complete the definition of the theory. After that I will be looking forward to many applications.

Two-Time Physics (2T-physics)

Starting with the [1998 formulation](#) of Two-Time Physics (2T-physics), which was inspired by 2T [notions](#) in earlier papers since 1995, evidence has been mounting that the ordinary formulation of physics, in a space-time with three space and one time dimensions (1T-physics), is insufficient to describe our world.

A one-page summary of the concepts of 2T-physics can be found in this [diagram](#) with related narrative below. Some of my lectures on 2T-physics at different levels (public, colloquium, research seminar) are available [online](#). For technical information please refer to my 2T-physics [papers](#).

According to the body of work in 2T-physics, there is more to space-time than can be garnered with 1T-physics. 2T-physics introduces additional one space and one time dimensions, which can coexist with the familiar 3+1 dimensions as well as extra space dimensions of tiny sizes known as [Kaluza-Klein-type dimensions](#), but the new ones have very different properties. First of all, the extra 1+1 dimensions in 2T-physics are not small. However, there are gauge symmetries that effectively reduce 2T-physics in 4+2 dimensions to 1T-physics in 3+1 dimensions without any Kaluza-Klein remnants. The reduction is not unique because there is an infinite variety of 3+1 embeddings in 4+2 dimensions (more generally $(d-1)+1$ in $d+2$), and this is what is non-trivial and rich in emergent space-times and 1T-physics content.

To help grasp the relation between 1T-physics and 2T-physics, consider the many possible shadows of a 3 dimensional object projected from different perspectives on the surrounding walls of a 3-dimensional room. A flatlander that can crawl and measure only on the surface of the walls would think that the shadows of different shapes are different “beasts” and move differently. Similarly, even though according to 2T-physics a unique dynamical system in 4+2 dimensions generates a large variety of 1-time “shadows”, 1T-physics presents these “shadows” in 3+1 dimensional space-times as different dynamical systems in terms of different Hamiltonians (different times).

In this way 1T-physics misses the underlying relationship between the “shadows” as well as the underlying properties (e.g. symmetries) of the higher dimensional space-time. Actually, it turns out that each “shadow” is a holographic image that retains all the information of the $d+2$ structure. This

information takes the form of hidden symmetries, dualities and other non-trivial structures, which are hard to notice by the 1T physicist that investigates the “shadows” (i.e. different dynamical systems). But he/she could in principle discover the hidden information. 2T-physics provides the missing information to the 1T physicist who can verify by experiment or computation that indeed the $d+2$ structure of space-time governs all levels of physics, from macroscopic to microscopic scales, in classical and quantum systems, including the fundamental physics of quarks, leptons and force particles described by the Standard Model of Particles and Forces, and beyond.

The permitted motions in $4+2$ phase space are highly symmetrical, as they are constrained by a $Sp(2,R)$ gauge symmetry that makes momentum and position *indistinguishable at any instant*. Such $Sp(2,R)$ symmetric motions in $4+2$ dimensions are completely compatible with the way physics is perceived in $3+1$ dimensions. In particular, there are no problems with causality or unitarity because the extra $1+1$ space-time (chosen in distinguishable ways from the point of view of 1T-physics) is removable by the gauge symmetry.

The two timelike dimensions were not introduced whimsically “by hand”. As mentioned above, 2T-physics is based on gauging the symplectic transformations $Sp(2,R)$ acting on phase space (X^M, P_M) . One of the fundamental results of this new gauge principle is that, in order to be nontrivial, it requires the theory to be formulated in a spacetime having at least two times. While taking exactly two timelike dimensions produces a coherent theory, investigations of alternatives with more than two times have been done (including alternatives to $Sp(2,R)$). So far such possibilities are ruled out because of problems with ghosts and unitarity, and this seems to confirm the special status of 2T-physics.

Recently, a field theoretic description of 2T-physics has been established. Amazingly, the best understood fundamental theory in Physics, the Standard Model of Particles and Forces (SM) in $3+1$ dimensions, is reproduced as one of the “shadows” of a parent field theory in $4+2$ dimensions. The emergent SM agrees with all aspects that actually work experimentally so far in the usual SM. It also suggests that the electro-weak phase transition must be driven by a new particle (perhaps the dilaton) thus leading to a new scenario for the mass

generation mechanism that is possibly related to other phase transitions in the history of the universe.

The field theoretic studies of 2T-physics have been generalized to supersymmetric field theory with $N=1,2,4$ supersymmetry. It is expected that the more constraining structure of the underlying $4+2$ theory has phenomenological consequences that would be relevant to distinguish 2T-physics from other approaches in experiments at the LHC starting in 2008, if supersymmetry is found experimentally at the TeV scale.

The field theory version of 2T-physics suggests new emergent principles in field theory. These have been enunciated in a paper that explores dual field theories in $(d-1)+1$ emergent spacetimes from a unifying field theory in $d+2$ spacetime.

Prior to recent success in field theory, the work on 2T-physics since 1998 had mostly concentrated on the worldline formalism of particles, and had demonstrated that 2T-physics stands above 1T-physics as a structure that encompasses and explains phenomena which appear very surprising from the point of view of 1T-physics.

The prior work on 2T-physics during 1998-2004 extended the initial concepts in several directions, including spinning particles, supersymmetry, and interactions of particles with background fields (electromagnetism, gravity, and all higher spin fields). Covariant quantization of 2T systems led to field theoretic equations of motion but without an action principle, and a non-commutative approach was developed for 2T-field theory in phase space. There was also some limited work on the world-sheet or world volume level for the 2T-physics formulation of strings and branes. Some hidden $10+2$ or $11+2$ structures in supergravity and M-theory, in the $AdS_5 \times S^5$ and other compactifications, were also identified and explained as features of 2T-physics.

After some excursions into String Field Theory during 2001-2004 to explore non-commutative aspects, extensive research on 2T-physics resumed in 2004-2005. This was sparked by the twistor superstring and its relation to the twistor gauge of 2T-physics. New unifying roles for twistors were discovered and a new approach to spinning particles led to a new hidden $SU(2,3)$ duality symmetry that includes conformal symmetry $SU(2,2)$.

These older results, along with the more recent field theory successes mentioned above, have established that 2T-physics is a structure that correctly describes, at least in principle, all the physics we have understood up to now. But 2T-physics emerged also as a unification scheme that suggests the existence of new relationships and new phenomena that are not even hinted by 1T-physics and which remain so far largely unexplored both theoretically and experimentally.

This 2T-physics point of view provides new mathematical tools and new insights for understanding our universe. It also suggests a new paradigm for the construction of a fundamental theory that is likely to impact on the quest for unification.

For systems that are already understood, 2T-physics tells us that the description of dynamics via the usual 1T-formalism should be interpreted as emergent dynamics that holographically represents an image of a deeper higher dimensional structure in one extra space and one extra time. A lot more work awaits to be done in this direction to reveal the hidden dimensions in various 1T systems, including in the field theory formalism. Ultimately we expect 2T-physics to be useful not only for insights into the deeper structures, but also as a calculational tool that takes advantage of the dualities and hidden symmetries in 1T-physics field theory.

For systems that are not yet understood or even constructed, such as M-theory, 2T-physics points to a possible approach for a more symmetric and more revealing formulation in 11+2 dimensions that can lead to deeper insights, including a better understanding of space and time. The 2T approach could be one of the possible avenues to construct the most symmetric version of the fundamental theory.

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Itzhak Bars

Department

Department of Physics and Astronomy

University of Southern California

Los Angeles, CA 90089-0484

Office

Seaver Science Center, SSC 216B

Tel: (213) 740-0047 Fax: (213) 740-6653

bars@usc.edu



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